

Optimizing the Josephson Parametric Amplifier

for Fast High-Fidelity Microwave Frequency Qubit Readout

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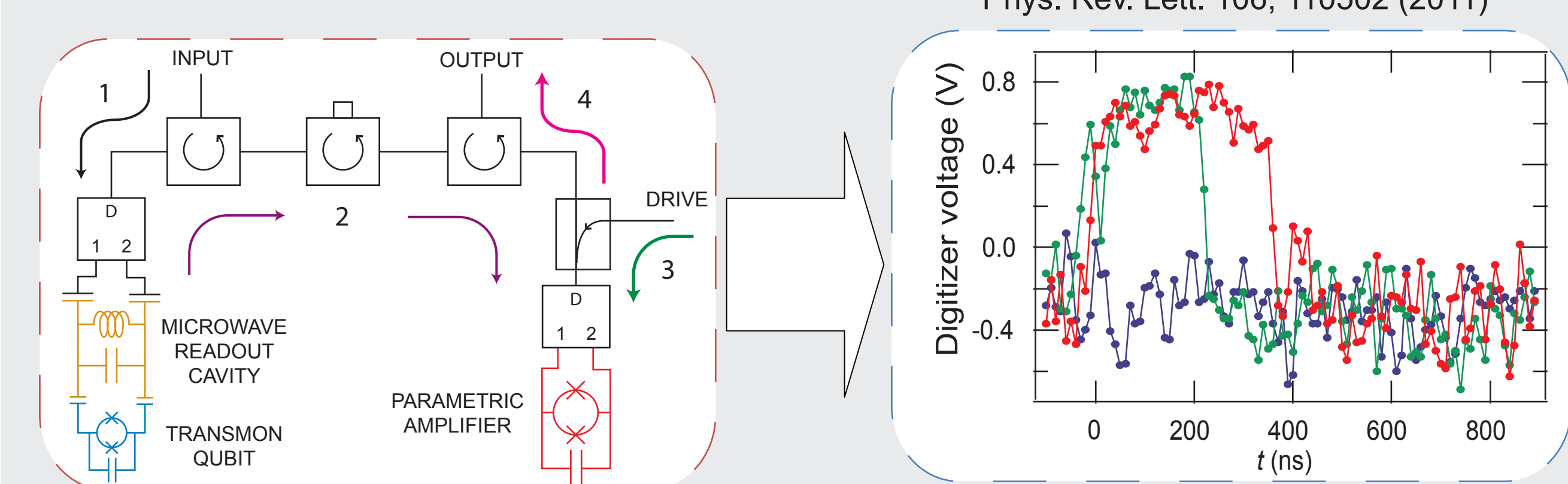


Introduction

Low noise microwave frequency amplifiers in demand:

- High fidelity readout for superconducting quantum bits
- Quantum optics at microwave frequencies
- Astrophysics: Axion detection

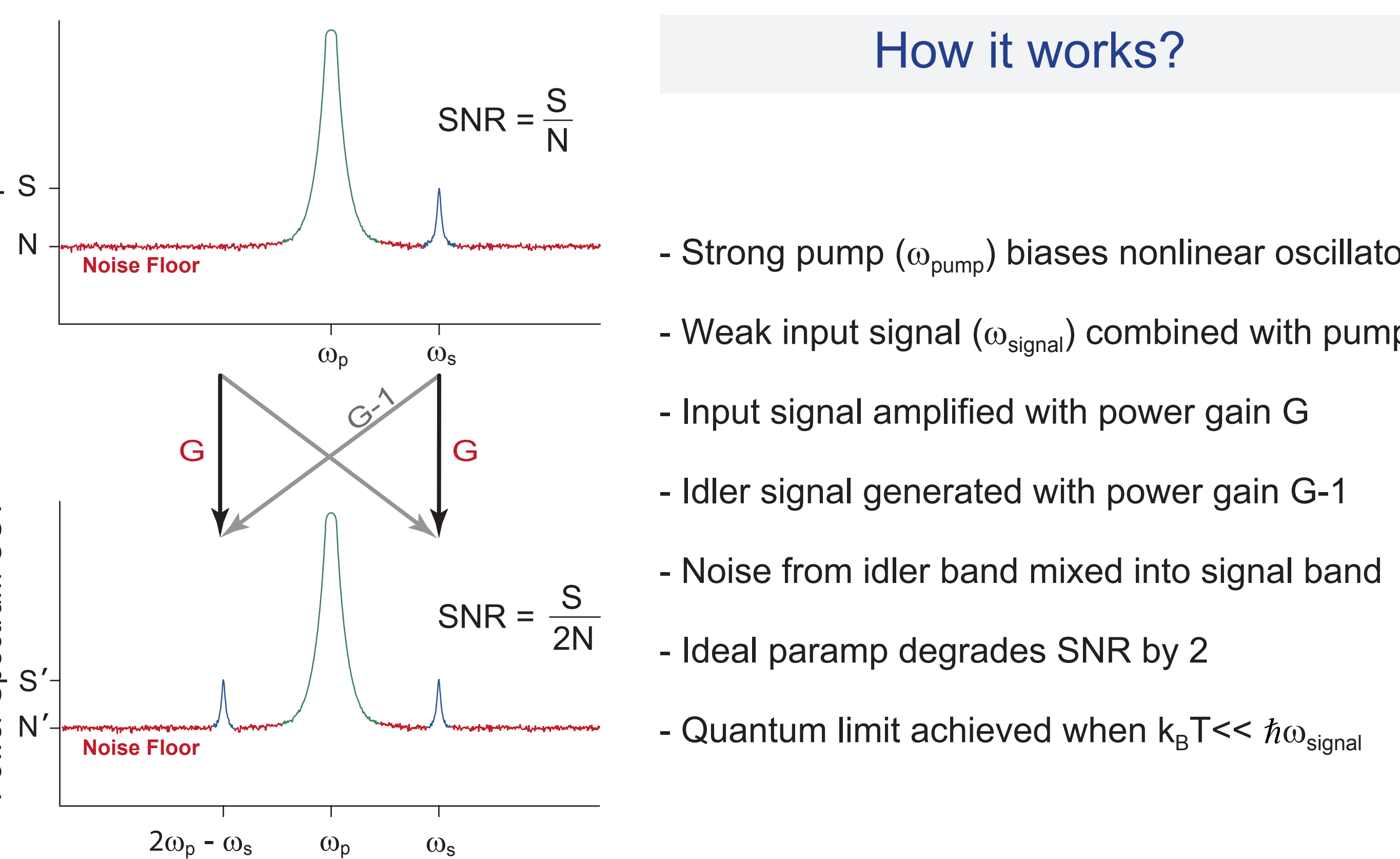
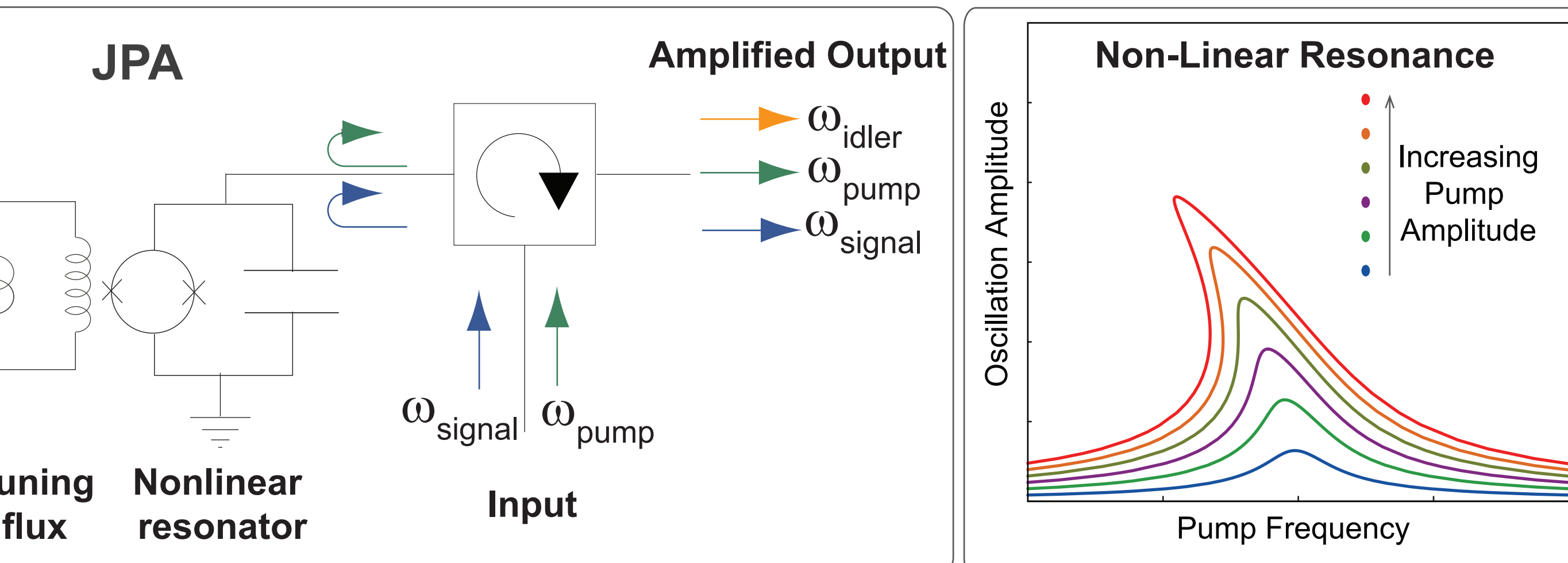
Motivation: Successful demonstration of high fidelity qubit readout and observation of quantum jumps using a Josephson parametric amplifier



Limitations: Amplifier saturation and narrow bandwidth

Goal: Explore device parameters to optimize parametric amplifier performance for bandwidth and dynamic range while maintaining high gain and low noise.

Josephson Parametric Amplifier

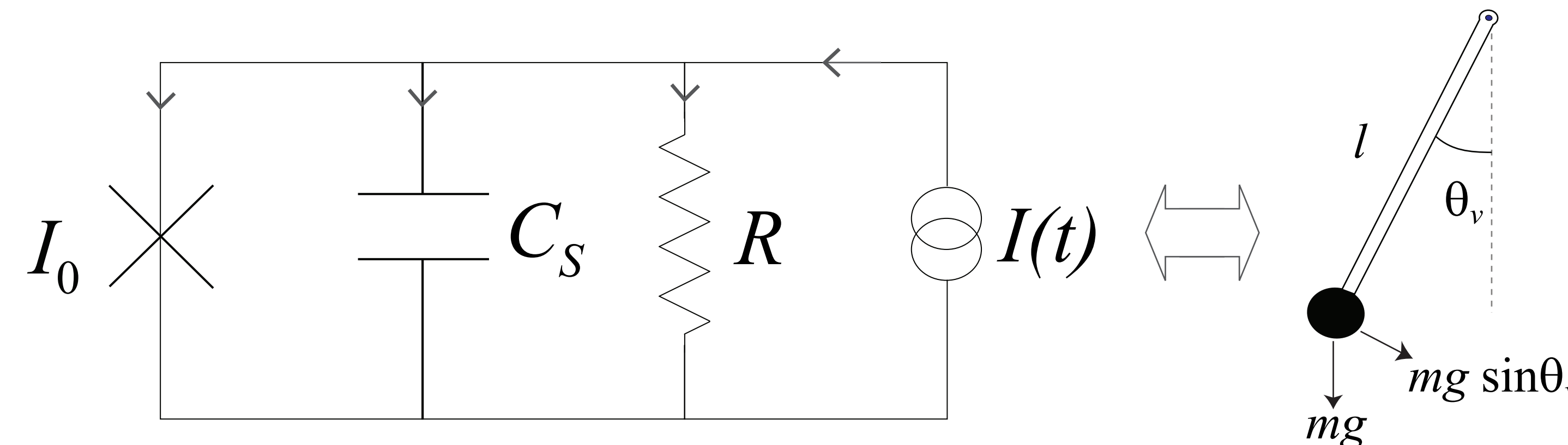


How it works?

- Strong pump (ω_{pump}) biases nonlinear oscillator
- Weak input signal (ω_{signal}) combined with pump
- Input signal amplified with power gain G
- Idler signal generated with power gain G-1
- Noise from idler band mixed into signal band
- Ideal paramp degrades SNR by 2
- Quantum limit achieved when $k_B T \ll \hbar \omega_{\text{signal}}$

Simulation Model

Numerically solve the differential equation for a driven, damped, nonlinear Josephson oscillator.



Differential equation

$$I_{\text{in}} + I_{\text{res}} + I_{\text{out}} = I_{\text{drive}}$$

$$C_s \varphi_0 \frac{d^2 \delta(t)}{dt^2} + \frac{\varphi_0}{R} \frac{d \delta(t)}{dt} + I_0 \sin(\delta(t)) = I(t)$$

Inertia Term
Damping Term
Restoring Force
Driving Term

$$I(t) = I_{\text{pump}} \cos(\omega_{\text{pump}} t) + I_{\text{noise}}(t)$$

Harmonic drive + noise

$$\omega_{\text{res}} = \sqrt{\frac{I_0}{C_s \varphi_0}}$$

Linear resonant frequency

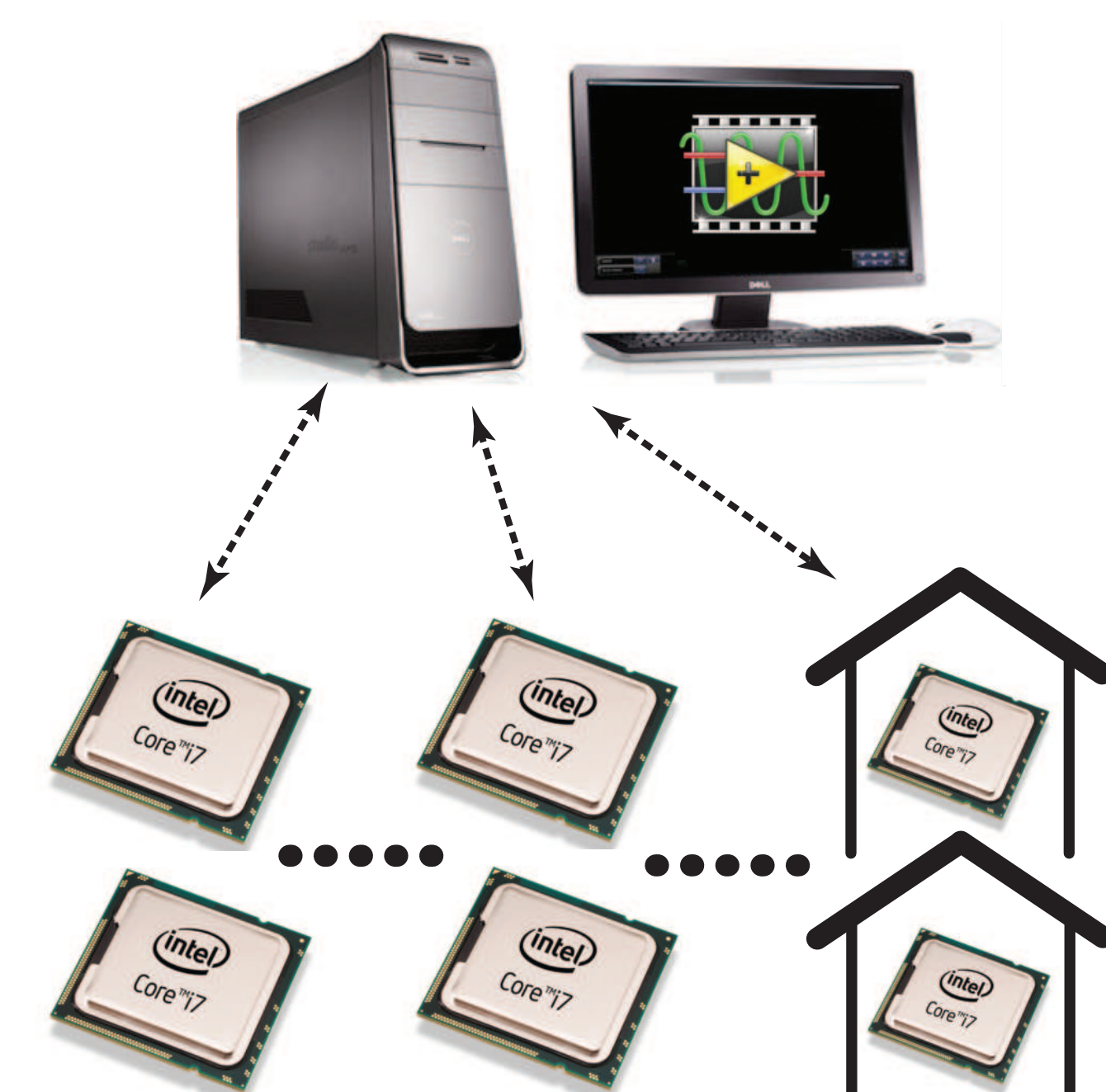
$$Q = R C_s \omega_{\text{res}}$$

Quality factor

$$\Omega = 2Q \frac{\omega_{\text{res}} - \omega_{\text{pump}}}{\omega_{\text{res}}}$$

Reduced drive detuning

Simulation architecture

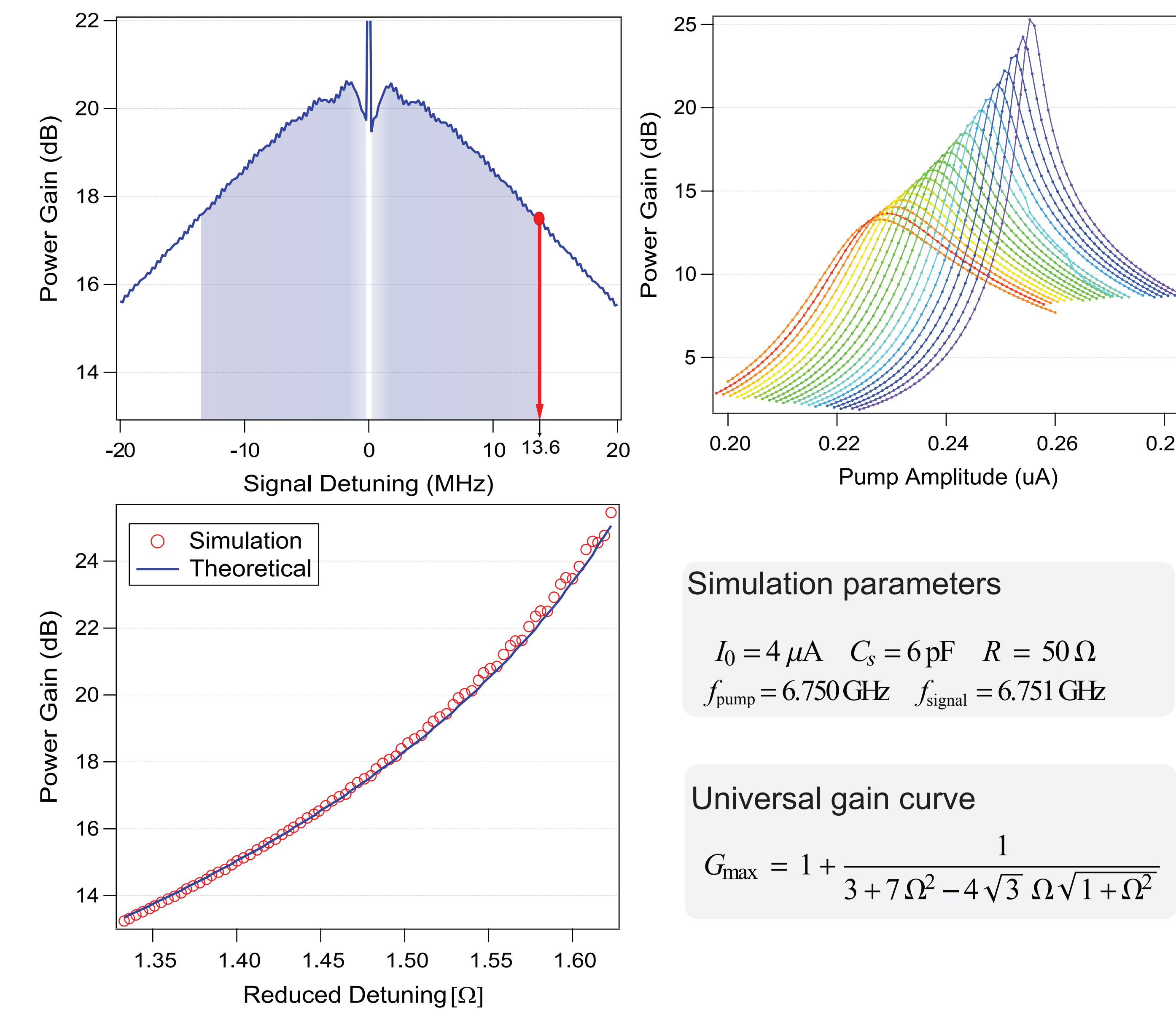


Numerical technique

- 4th order Runge-Kutta algorithm
- Finite temperature effects included using gaussian noise
- Solver written in C++ and compiled as DLL
- Labview front end for user friendly interface
- Labview program controls solver across multiple cores on multiple computers
- Ability to extend to any number of computers over a network

Numerical Simulation Results

Noise-free simulation & theory



Simulation parameters

$$I_0 = 4 \mu\text{A} \quad C_s = 6 \text{ pF} \quad R = 50 \Omega$$

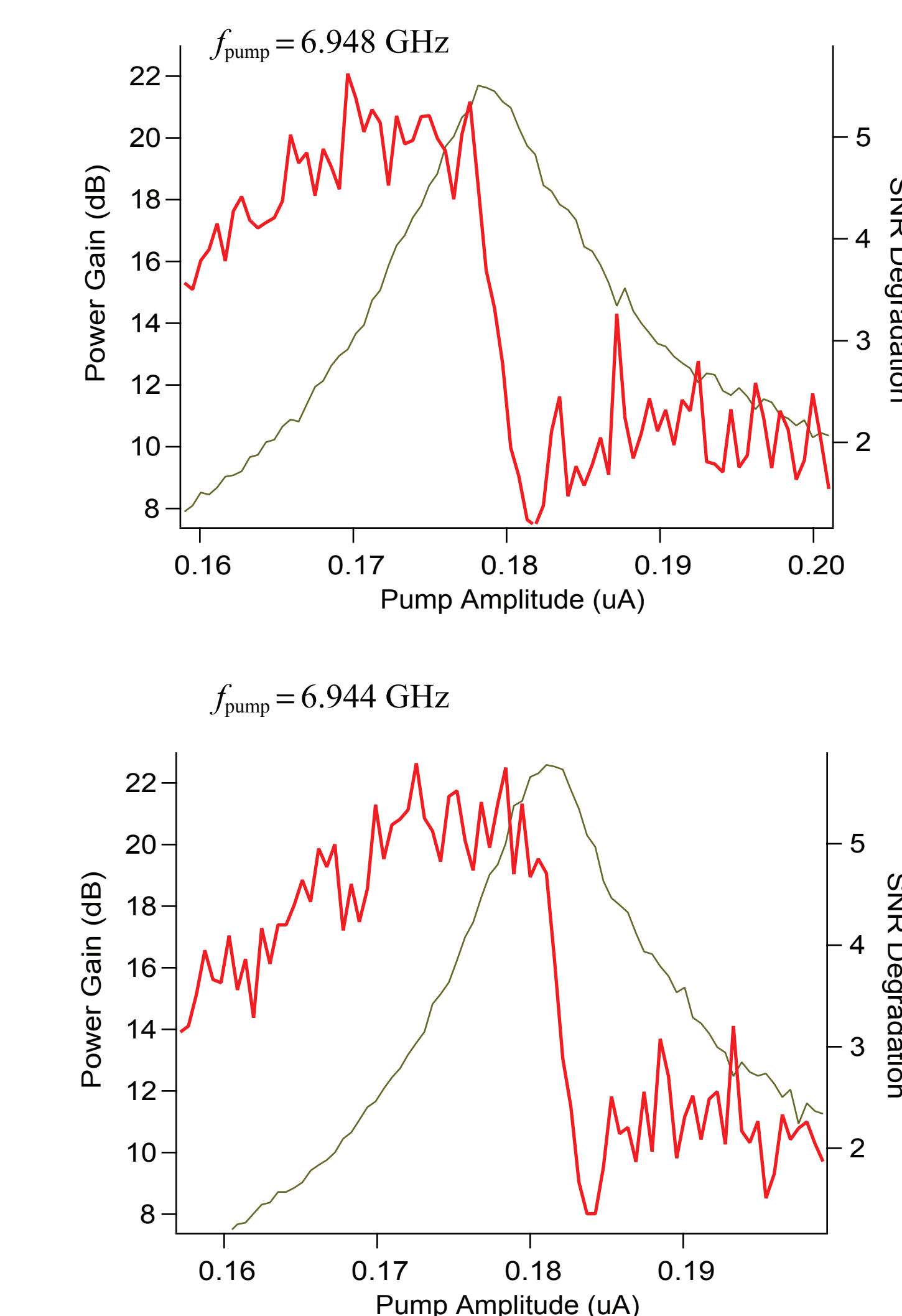
$$f_{\text{pump}} = 6.750 \text{ GHz} \quad f_{\text{signal}} = 6.751 \text{ GHz}$$

Universal gain curve

$$G_{\text{max}} = 1 + \frac{1}{3 + 7\Omega^2 - 4\sqrt{3}\Omega\sqrt{1 + \Omega^2}}$$

Simulation agrees with ideal noise-free analytical theory

Noise performance



Simulation parameters

$$I_0 = 8 \mu\text{A} \quad C_s = 12 \text{ pF} \quad R = 50 \Omega$$

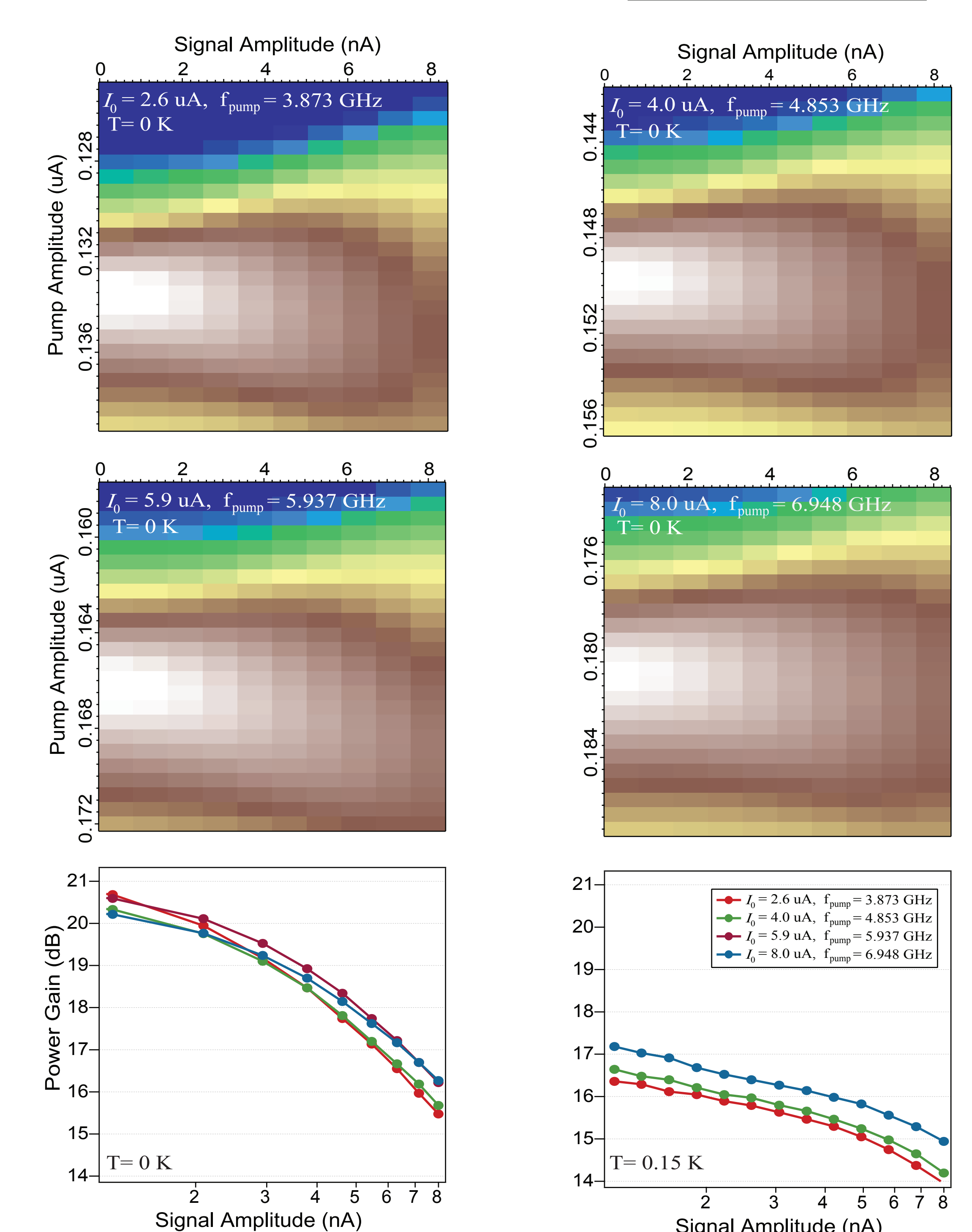
$$f_{\text{signal}} = f_{\text{pump}} + 1 \text{ MHz} \quad T = 0.15 \text{ K}$$

Noise Results

- Simulate with and without noise to compute SNR degradation
- SNR degradation shows asymmetry around maximum gain point
- Ideal SNR degradation of 2 observed

Dynamic range

Extract gain as a function of pump and signal amplitude



- Amplifier frequency band tunable with critical current
- No significant change in 1 dB compression point across 4 - 7 GHz band
- Noise suppresses maximum gain for given drive detuning (Ω)

Conclusions & Future Directions

- Implemented numerical solver for studying parametric amplifier performance
- Noise free simulations agree with predictions of analytical theory
- Simulations with noise show gain suppression and asymmetry in SNR degradation
- Explore maximum achievable bandwidth by varying quality factor (Q)
- Study the effects of stray parameters on amplifier performance
- Simulate multiple junction paramp for improved dynamic range

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